Classification of Malignant Melanoma in Canines

Week 1–2 Progress – Foundation Phase

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Where we are in the plan

Phase	Weeks	Now
Foundation	1–3	← we are here (W2)
Model Development	3–8	
Evaluation & Analysis	8–10	
Documentation	8–12	

12-week project: 26 May → 14 Aug 2026, submission 17 Aug 2026.

Planned vs. done (Gantt)

Task	Weeks	Status
Environment Setup & Framework	W1	Done
Literature Review & Refinement	W1–2	Done
Data Acquisition & Preprocessing	W1–3	~60%

On schedule.

1 · Environment setup ✓

- Isolated virtual environment (.venv) – other projects untouched
- Python 3.12 + OpenCV, scikit-image, scikit-learn, OpenSlide (PyTorch next)
- Reproducible: requirements.txt + requirements-lock.txt + setup_env scripts
- Clean modular src/ package, version-controlled (Git)

2 · Literature review refined ✓

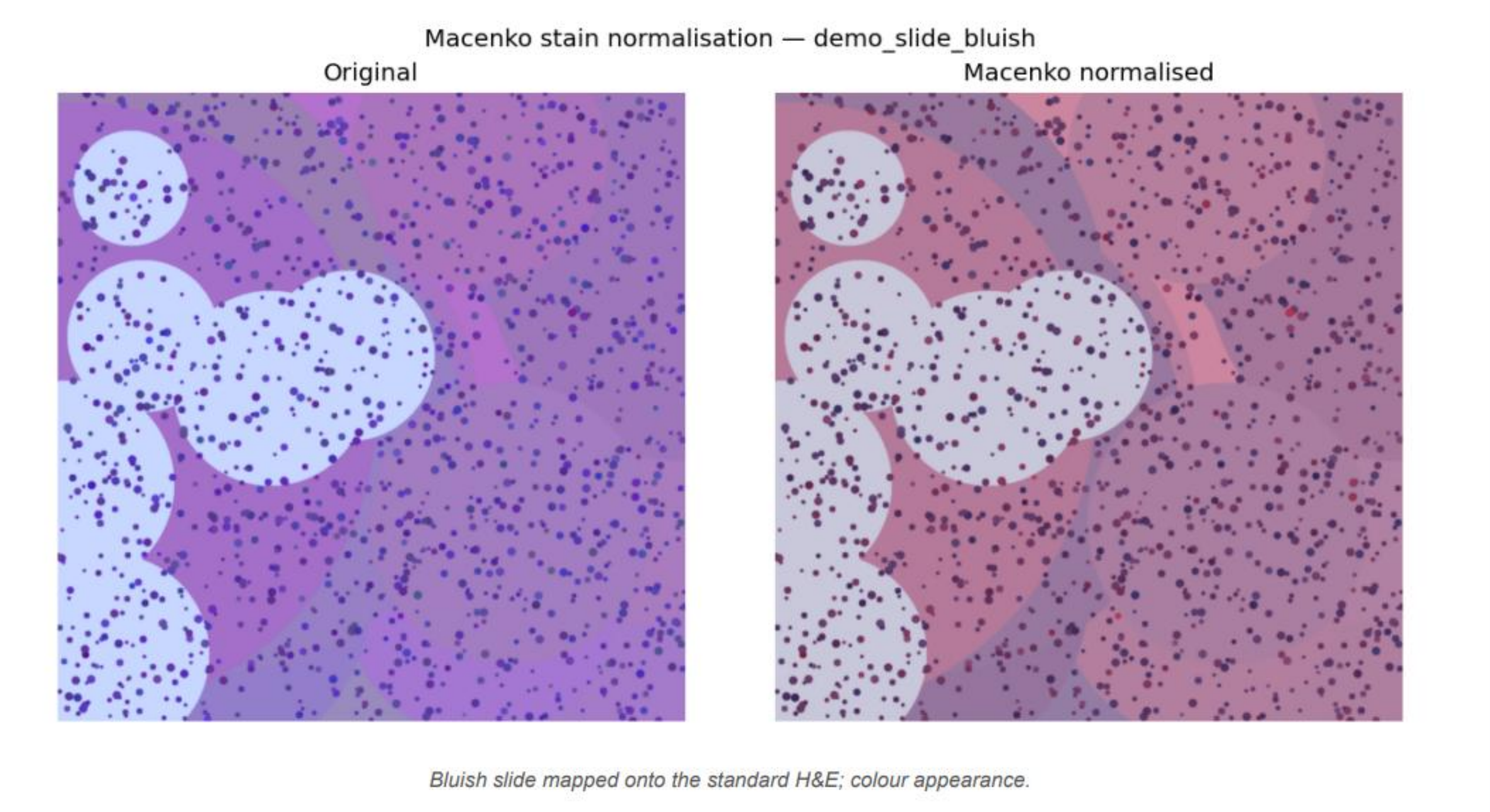
- Segmentation: U-Net (Ronneberger 2015), Attention U-Net (Oktay 2018)
- Classification: ResNet-50 (He 2016), EfficientNet-B3 (Tan & Le 2019)
- Preprocessing: Macenko stain normalisation (2009)
- Interpretability: Grad-CAM (Selvaraju 2017)
- Comparative oncology: Gillard (2014), Prouteau & André (2019)

3 · Preprocessing pipeline – built & validated ✓

Whole-slide image → QA → Macenko normalise → patch extraction → 70/15/15 split

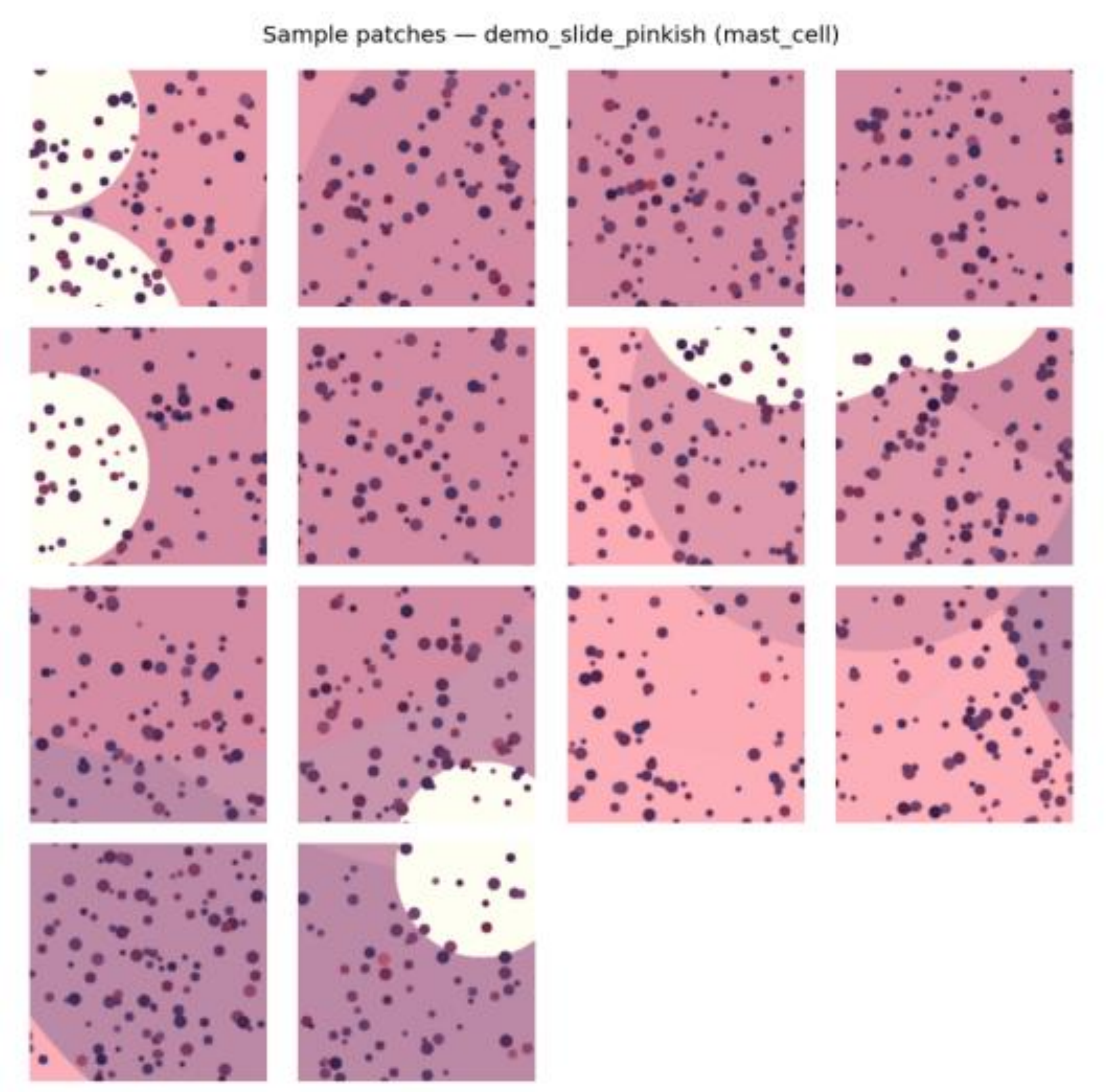
- Quality check flags blurry / low-tissue / washed-out slides automatically
- Macenko removes scanner/lab colour variation
- Only tissue patches (≥50%) are kept
- Stratified split preserves class balance

Result – stain normalisation (before / after)



Bluish slide mapped onto the standard H&E; colour appearance.

Result – extracted tissue patches



256×256 patches; background tiles automatically discarded.

Validation run (synthetic slides)

Metric	Value
Slides processed	2
Passed QA	2 / 2
Patches extracted	26
Split (train/val/test)	18 / 4 / 4

Pipeline runs end-to-end and is ready to run unchanged on the real CATCH slides once the TCIA download completes.

Risks & mitigations

Risk	Mitigation
TCIA download / large files	Pipeline validated on synthetic data; NBIA workflow documented
GPU for training	Colab Pro + university HPC fallback
Class imbalance	Stratified split now; weighted loss + augmentation next

Next fortnight (Weeks 3–5)

- Complete CATCH download → run pipeline on all real slides
- Finalise patch dataset + class statistics
- Start U-Net segmentation (baseline → ResNet-34 encoder)

Questions: QA thresholds OK? · final class list? · magnification priority?

Thank You

Questions & feedback welcome